

① Horner's Method

⇒ if x_0 is Not given do vector of constant value

⇒ rewrite $f(x)$ in decending order (higher to lower)

Example 2 Use Horner's method to evaluate $P(x) = 2x^4 - 3x^2 + 3x - 4$ at $x_0 = -2$.

$$P(x) = 2x^4 + 0 - 3x^2 + 3x - 4$$

$\downarrow$ $\downarrow$
 x^3 is missing factoring -2×2

$$x_0 = -2 \quad \begin{array}{r|rrrrr} 2 & 0 & -3 & 3 & -4 \\ \downarrow \textcircled{1} & & -4 \textcircled{2} & 8 \textcircled{3} & -10 & 14 \\ \hline 2 & -4 & 5 & -7 & \boxed{10} \end{array} = P(-2)$$

$\Rightarrow$

$$\begin{array}{r|rrrr} -2 & 2 & -4 & 5 & -7 \\ \downarrow & & -4 & 16 & -42 \\ \hline 2 & -8 & 21 & \boxed{-49} \end{array} = P'(-2)$$

$$x_1 = x_0 - \frac{P(x)}{P'(x)}$$

$$x_1 = -2 - \frac{10}{-49} = -1.796$$

$$-1.796 \quad \begin{array}{r|rrrrr} 2 & 0 & -3 & 3 & 4 \\ \downarrow & & -3.592 & 6.451 & -6.197 & 5.741 \\ \hline 2 & -3.592 & 3.451 & -3.197 & \boxed{9.741} \end{array} = P(x_1)$$

$\Rightarrow$

$$-1.796 \quad \begin{array}{r|rrrr} 2 & -3.592 & 3.451 & -3.197 \\ \downarrow & & -3.592 & 12.902 & -29.369 \\ \hline 2 & -7.184 & 16.353 & \boxed{-32.566} \end{array} = P'(x_1)$$

$$x_2 = x_1 - \frac{P(x_1)}{P'(x_1)} = -1.4418$$

Q 18

1. Find the approximations to within 10^{-4} to all the real zeros of the following polynomials using Newton's method.

a. $f(x) = x^3 - 2x^2 - 5$ $x_0 \Rightarrow$ is not given so use Constant Vector

$$= x^3 - 2x^2 + \underbrace{0}_{x \text{ is missing}} - \underbrace{5}_{(5x-1)}$$

$$\begin{array}{r|rrrr} -1 & 1 & -2 & 0 & -5 \\ & \downarrow & -1 & 3 & -3 & + \\ \hline & 1 & -3 & 3 & -8 & \Rightarrow P(x) \end{array}$$

$$\begin{array}{r|rrrr} -1 & 1 & -3 & 3 & \\ & \downarrow & -1 & 4 & + \\ \hline & 1 & -4 & 7 & \Rightarrow P'(x) \end{array}$$

$$x_1 = x_0 - \frac{P(x)}{P'(x)} = -1 - \frac{-8}{7}$$

$$x_1 = 0.1428 \approx 0.143$$

$$\begin{array}{r|rrrr} 0.143 & 1 & -2 & 0 & -5 \\ & \downarrow & 0.143 & -0.265 & -0.037 & + \\ \hline & 1 & -1.857 & -0.265 & -5.037 & \Rightarrow P(x) \end{array}$$

$$\begin{array}{r|rrrr} 0.143 & 1 & -1.857 & -0.265 & \\ & \downarrow & 0.143 & -0.245 & + \\ \hline & 1 & -1.714 & -0.51 & \Rightarrow P'(x) \end{array}$$

$$x_2 = x_1 - \frac{P(x)}{P'(x)}$$

$$x_2 = -9.7335$$

② Muller's Method

→ p_0, p_1, p_2 are given to find p_3

→ just memorise "C" formula.

$$a = \frac{(p_1 - p_2)[f(p_0) - f(p_2)] - (p_0 - p_2)[f(p_1) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)}, \rightarrow \text{given}$$

$$b = \frac{(p_0 - p_2)^2[f(p_1) - f(p_2)] - (p_1 - p_2)^2[f(p_0) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)}, \rightarrow \text{given}$$

$$C = p_3 - p_2 = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}, \rightarrow \text{memorize}$$

↑
choose sign depend on "b" sign

Q 30 3. Repeat Exercise 1 using Müller's method.

a. $f(x) = x^3 - 2x^2 - 5$

by choosing $p_0 = 0, p_1 = 1, p_2 = 2 \Rightarrow$ Can be any number

$$f(p_0) = -5$$

$$f(p_1) = -6$$

$$f(p_2) = -5 = C$$

$$a = \frac{(p_1 - p_2)[f(p_0) - f(p_2)] - (p_0 - p_2)[f(p_1) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)},$$

$$b = \frac{(p_0 - p_2)^2[f(p_1) - f(p_2)] - (p_1 - p_2)^2[f(p_0) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)},$$

$$c = \frac{-2c}{b \pm \sqrt{b^2 - 4ac}}$$

$$2 + \sqrt{(1) - 4(1)(-5)}$$

$$a = \frac{(1-2)[-5+5] - (-5+5)[-6+5]}{(0-2)(1-2)(0-1)} = 1$$

$$b = \frac{(0-2)^2[-6+5] - (1-2)^2[-5+5]}{(0-2)(1-2)(0-1)} = 2$$

$$P_3 = P_2 - \frac{-2c}{1} = 2 - \frac{-2(-5)}{1} = 0.4808378$$